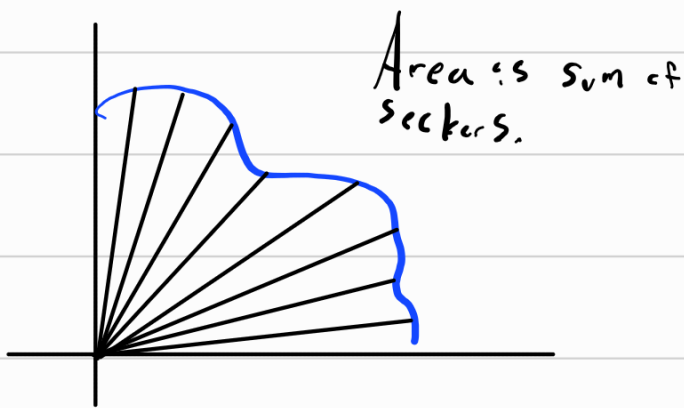


Area:

Really, the area of a sector of a circle is

$$A = \frac{1}{2} r^2 \theta$$

This is important as when calculating area in polar we will vary radius as θ increases



So, if $r = f(\theta)$, the area will be

$$A = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta = \frac{1}{2} \int_a^b r^2 d\theta$$

Ex: Area in one loop of the four-leaved rose
 $r = c \cos 2\theta$

Bounds, $[0, \pi/2]$

$$\begin{aligned} A &= \frac{1}{2} \int_0^{\pi/2} r^2 d\theta = \frac{1}{2} \int_0^{\pi/2} c^2 \cos^2(2\theta) d\theta \\ &= \frac{1}{4} \int_0^{\pi/2} (1 + \cos(4\theta)) d\theta \\ &= \frac{1}{4} \left[\theta + \frac{1}{4} \sin(4\theta) \right]_0^{\pi/2} \\ &= \frac{1}{4} \left[\left(\frac{\pi}{2} + 0 \right) - (0 + 0) \right] \end{aligned}$$

$$A = \frac{\pi}{8}$$

Ex: Area inside the circle $r = 3 \sin \theta$ but
outside $r = 1 + \sin \theta$

First, find intersection points

$$3\sin\theta = 1 + \sin\theta$$

$$2\sin\theta = 1$$

$$\sin\theta = \frac{1}{2} \rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$A = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3\sin\theta)^2 - (1 + \sin\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 9\sin^2\theta - 1 - 2\sin\theta - \sin^2\theta d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 8\sin^2\theta - 2\sin\theta - 1 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 4 - 4\cos(2\theta) - 2\sin\theta - 1 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 3 - 4\cos(2\theta) - 2\sin\theta d\theta$$

$$= \frac{1}{2} \left[3\theta - 2\sin(2\theta) + 2\cos\theta \right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

$$= \frac{1}{2} \left[\left(\frac{5\pi}{2} - 2\left(-\frac{\sqrt{3}}{2}\right) + 2\frac{-\sqrt{3}}{2} \right) - \left(\frac{\pi}{2} - 2\frac{\sqrt{3}}{2} + 2\frac{\sqrt{3}}{2} \right) \right]$$

$$= \frac{1}{2} 2\pi$$

$$= \pi$$

Arc length

In Cartesian Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta$$

We have $r = f(\theta)$, so, to find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ we use chain and product rule

$$\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta, \quad \frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta$$

Then

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= \left(\frac{dr}{d\theta} \cos \theta - r \sin \theta\right)^2 + \left(\frac{dr}{d\theta} \sin \theta + r \cos \theta\right)^2 \\ &= \underbrace{\left(\frac{dr}{d\theta}\right)^2 \cos^2 \theta} + \underbrace{-2r \frac{dr}{d\theta} \cos \theta \sin \theta} + \underbrace{r^2 \sin^2 \theta} + \underbrace{\left(\frac{dr}{d\theta}\right)^2 \sin^2 \theta} + \underbrace{2r \frac{dr}{d\theta} \cos \theta \sin \theta} + \underbrace{r^2 \cos^2 \theta} \\ &= \left(\frac{dr}{d\theta}\right)^2 + r^2 \end{aligned}$$

So arc length,

Arc length

$$L = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Ex: Arc length of
 $r = e^{\theta/2}$ on $0 \leq \theta \leq 2\pi$

$$r^2 = e^{\theta}$$

$$\frac{dr}{d\theta} = \frac{1}{2} e^{\theta/2}$$

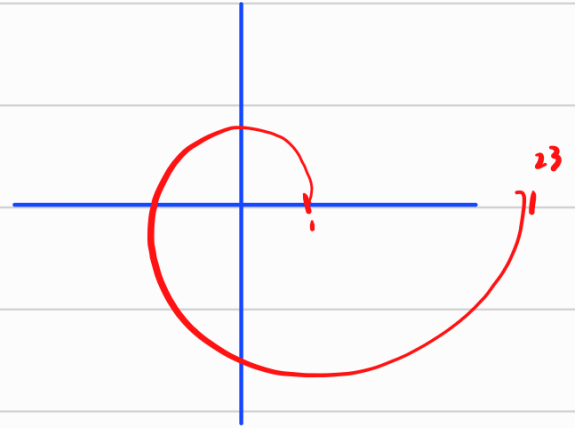
$$L = \int_0^{2\pi} \sqrt{e^{\theta} + \left(\frac{1}{2}e^{\theta/2}\right)^2} d\theta$$

$$= \int_0^{2\pi} \sqrt{\frac{5}{4}e^{\theta}} d\theta$$

$$= \frac{\sqrt{5}}{2} \int_0^{2\pi} e^{\theta/2} d\theta$$

$$= \sqrt{5} \left[e^{\theta/2} \right]_0^{2\pi}$$

$$= \sqrt{5}(e^{\pi} - 1)$$



Tangents

We saw that

$$\frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta - r \sin \theta, \quad \frac{dy}{d\theta} = \frac{dr}{d\theta} \sin \theta + r \cos \theta$$

$$\text{So, } \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

$$\text{Note, if } r=0, \quad \frac{dy}{dx} = \tan \theta$$

If we are looking for horizontal tangents,
check when $\frac{dy}{d\theta} = 0$

If we are looking for vertical tangents
check when $\frac{dx}{d\theta} = 0$

Ex: Find Horizontal and vertical tangents of

$$r = 1 + \sin \theta$$

$$\begin{aligned}\frac{dx}{d\theta} &= \cos \theta - (1 + \sin \theta) \sin \theta \\ &= \cos^2 \theta - \sin^2 \theta - \sin \theta \\ &= (1 - \sin^2 \theta) - \sin^2 \theta - \sin \theta \\ &= 1 - 2\sin^2 \theta - \sin \theta \\ &= (1 + \sin \theta)(1 - 2\sin \theta)\end{aligned}$$

$$\begin{aligned}\frac{dy}{d\theta} &= \cos \theta \sin \theta + (1 + \sin \theta) \cos \theta \\ &= \cos \theta (\sin \theta + 1 + \sin \theta) \\ &= \cos \theta (1 + 2\sin \theta)\end{aligned}$$

$$\frac{dx}{d\theta} = 0 \text{ if } \sin \theta = -1 \text{ or } \frac{1}{2}$$

$$\text{So } \theta = -\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$$

Vertical tangent

$$\frac{dy}{d\theta} = 0 \text{ if } \cos \theta = 0$$

$$\text{or } \sin \theta = -\frac{1}{2}$$

$$\text{So, } \theta = \pm \frac{\pi}{2}, \frac{-\pi}{6}, \frac{-5\pi}{6}$$

Horizontal tangent

Notice $-\frac{\pi}{2}$ is in both, so we need to check

$$\begin{aligned}\lim_{\theta \rightarrow -\frac{\pi}{2}} \frac{dy}{dx} &= \lim_{\theta \rightarrow -\frac{\pi}{2}} \frac{\cos \theta (1 + 2\sin \theta)}{(1 + \sin \theta)(1 - 2\sin \theta)} \stackrel{\text{L'H}}{=} \lim_{\theta \rightarrow -\frac{\pi}{2}} \frac{-\sin \theta (1 + 2\sin \theta) + 2\cos^2 \theta}{\cos \theta (1 - 2\sin \theta) - 2\cos \theta (1 + \sin \theta)} \\ &= \infty\end{aligned}$$

So $\frac{\pi}{2}$ is a vertical tangent.

So, $r = 1 + \sin \theta$ has vertical tangent when

$$\theta = -\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6} \quad \text{i.e.} \quad \left(\frac{3}{2}, \frac{\pi}{6}\right), \left(\frac{3}{2}, \frac{5\pi}{6}\right) \quad \text{and the Pole,}$$

$\theta = -\frac{\pi}{2}, r = 0$

horizontal tangent at

$$\theta = -\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{2}, \quad \text{i.e.} \quad \left(2, \frac{\pi}{2}\right), \left(\frac{1}{2}, -\frac{\pi}{6}\right), \left(\frac{1}{2}, -\frac{5\pi}{6}\right)$$